Sachi Patel

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Education

Columbia University Expected May 2028
Accelerated M.S. (4+1) in Computer Science, Machine Learning Pathway *New York, NY*

Barnard College of Columbia University Expected May 2027
B.A. in Computer Science & Economics | GPA: 3.83 | New Pathways Bridgewater Scholar *New York, NY*
– **Selected Coursework:** Data Structures, Computer Architecture, Systems Programming, Natural Language Processing, Artificial Intelligence, Applied Machine Learning, Databases, Agentic Engineering, Analysis of Algorithms, Discrete Math, Linear Algebra

Skills

Languages/Frameworks: Python, Java, JavaScript, TypeScript, C, SQL, Bash, R; React, Node.js, Flask, FastAPI, Next.js
ML/AI: PyTorch, scikit-learn, Hugging Face, LangChain, Google ADK, MCP, RAG, FAISS, Pandas, NumPy
Cloud/Infra: AWS (CDK, ECS Fargate, EC2, S3, Lambda, CloudWatch), Terraform, Docker, GitLab CI/CD
Observability/Data: OpenTelemetry (OTLP), Prometheus, Grafana, PostgreSQL, BigQuery

Work Experience

Columbia Build Lab Sep. 2026 – Present
Software Engineer – Kernel *New York, NY*
– Building multi-agent LLM system to detect and resolve conflicting customer records across pharma data sources
– Developing web-scraping agent that parses hospital webpages for evidence of clinic-to-health-system ownership
– Building source-agnostic data pipeline (Supabase, AWS S3, Doppler) feeding a human-in-the-loop review queue

Goldman Sachs Jun. 2026 – Aug. 2026
Software Engineering Intern – Data & AI Solutions Engineering *New York, NY*
– Architected SRE observability infrastructure (AWS CDK, Terraform) for a Python data/AI platform serving 15+ teams
– Deployed OTel Collector, Prometheus, Grafana, and CloudWatch Exporter services on ECS Fargate for metric aggregation
– Collected 99 application and infrastructure metrics; ingested into a lakehouse DB for long-term analytics
– Built MCP tooling querying lakehouse data; agent skills assisting custom metric creation, dashboards, and alerting

Depository Trust and Clearing Corporation Jun. 2025 – Aug. 2025
Software Engineering Intern – Cloud Center for Enablement *Jersey City, NJ*
– Built internal automation to detect idle EC2/EBS volumes across 25+ SYSIDs; alerts drove $350K in annualized savings
– Developed MVP cost-savings dashboard (Python, Flask, REST APIs) integrating CloudAware and AWS Cost Explorer data
– Presented cost insights to CloudOps teams and SYSID owners; drove decommissioning of 200+ idle resources org-wide

Rajiv Sethi Lab, Barnard College Jan. 2024 – Present
Research Assistant *New York, NY*
– Built data pipeline (Python, BeautifulSoup) scraping price data/probabilities from prediction markets/models across 41 events
– Implemented profitability test (Python/MATLAB) analyzing trading strategies; extracted insights on model vs. market accuracy
– Co-authored paper finding FiveThirtyEight beat Polymarket by 17.9% on popular vote; published in ACM CI '25 proceedings

Barnard Computational Science Center (CSC) Aug. 2024 – Present
Senior Computing Fellow *New York, NY*
– Mentored 120+ students in Python, R, and MATLAB across 3 data science courses; debugged code and guided final projects
– Led 30+ office hours sessions supporting Coding Markets, Privacy in a Networked World, and Intro to Data Science
– Designed and led 3 workshops with modular Colab exercises covering data wrangling, visualization, and scripting

Projects

Agentic Stock Prediction System Jun. 2026 (In Progress)
– Building autonomous trading agent (Google ADK) that orchestrates tool-calling across market data APIs/technical indicators
– Integrating LLM reasoning over real-time and historical market data to generate buy/sell/hold recommendations with rationale
– Designing evaluation harness that backtests agent recommendations against historical returns to measure decision quality

RAG-Powered Codebase Assistant Jan. 2026
– Built RAG assistant that ingests GitHub repos, parses Python code with tree-sitter, answers questions via natural language queries
– Implemented semantic search using sentence-transformers/FAISS; integrated LangChain + Ollama for conversational UI
– Engineered multi-repo indexing with chunked parsing; achieved sub-second query latency across 10K+ lines of code

HTTP Web Server Dec. 2024
– Built HTTP/1.0 server in C using POSIX sockets; forked child processes to handle concurrent connections and serve static files
– Implemented request parsing, URI validation, directory traversal prevention, and error handling (200/301/400/404/501)

Activities & Leadership

Columbia Multicultural Business Association (Conference Lead) | Society of Women Engineers (Committee Member) | Columbia Women in Computer Science (Event Coordinator) | Grace Hopper Conference '24 & '25 | Columbia DJ Collective